Credit Risk Intelligence & Talk-to-Data Platform

From risk signals to decisions — automated underwriting, explainable AI, conversational analytics

Introduction

Unified Credit Platform

AutoML scoring + deterministic rules + conversational SQL in one production UI

Scale & Data

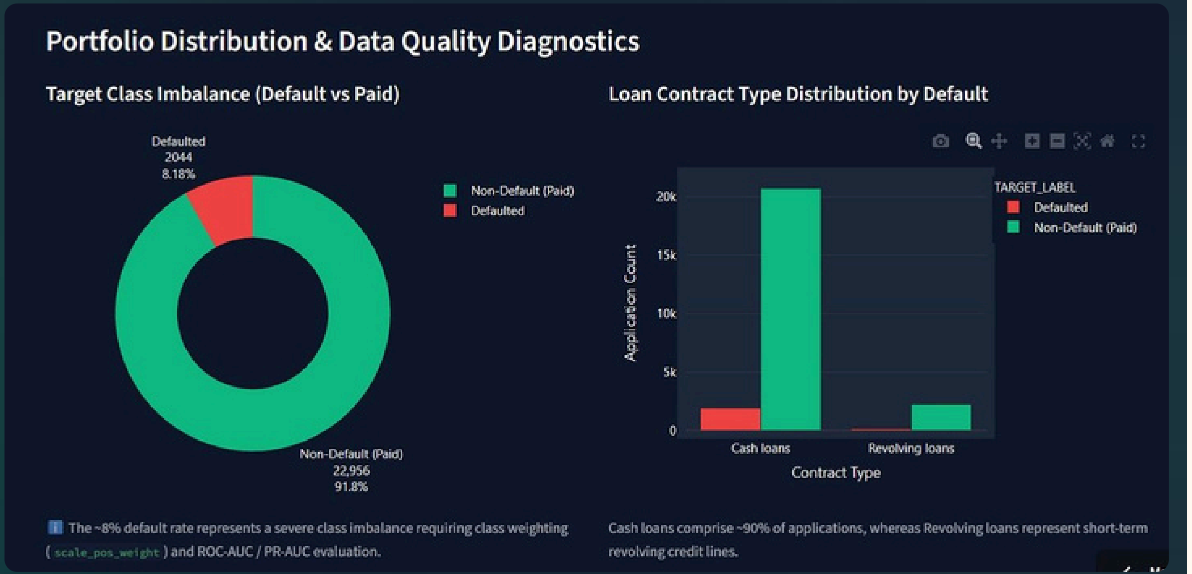
307,511 loans • 6 relational tables • 460+ engineered features

Performance

XGBoost champion — ROC-AUC 0.778 • PR-AUC 0.284 • 8.07% severe imbalance

Enterprise Ready

Dockerized deployment • Streamlit Community Cloud hosting • zero-maintenance



Business Problem & Dataset

An a to my

Challenge

Detect likely defaulters among thin-file borrowers — reduce false negatives

Data Scope

- 307,511 loan records (primary table)
- Telemetry: bureau, POS, cards, installment history
- 122 baseline features + 460+ engineered

Class Imbalance

91.93% Non-Default vs 8.07% Default

Strategy

Optimize ROC-AUC & PR-AUC • calibrated thresholding (scale_pos_weight=11.4)

End-to-End System Architecture

Data & Feature Engineering

Polars/Pandas batch aggregations → Parquet

Model Inference

Scikit-learn preprocess + FLAML AutoML (XGBoost)

Explainability

Real-time TreeSHAP • per-applicant waterfall

Conversational SQL

Gemini client → safe read-only SQLite queries

Serving

Streamlit dashboard • Docker & Docker Compose

```
Raw CSV files  -->  DataLoader  -->  Merged Parquet Cache (307k x 467 features)
                        |                          |
                        v                          v
                     SQLite DB          DataPreprocessor
                  (credit_risk.db)      (clean, impute, encode)
                        |                          |
                        |                          v
           Talk-to-Data |                FLAML AutoML training
           (Gemini LLM) |                -->  XGBoost / LightGBM model
                        |                          |
                        v                          v
                   Query Runner            RiskPredictor (scores 0-1000)
                        |                  TreeSHAP (local explanations)
                        |                  RuleDeriver (mined IF-THEN rules)
                        |                  BusinessRulesEngine (hard policies)
                        |                          |
                        +-------------+------------+
                                      v
                           Streamlit Web UI (app.py)
```

Module 1 — Interactive EDA & Portfolio Profiling

Key signals

- Age cohorts: 18–25 ≈ 12.3% default vs 55–65 ≈ 5.4%
- EXT_SOURCE_2 & EXT_SOURCE_3 — strongest separation
- Late payment (DPD>30) → ~3.8× default likelihood
- Revolver utilization >80% → severe liquidity stress
- Missing-value slider flags features >40% sparsity for imputation

Module 2 — Machine Learning & AutoML

Model Exploration

LightGBM, RF, LR, ExtraTrees, XGBoost — 3-fold CV

Why XGBoost

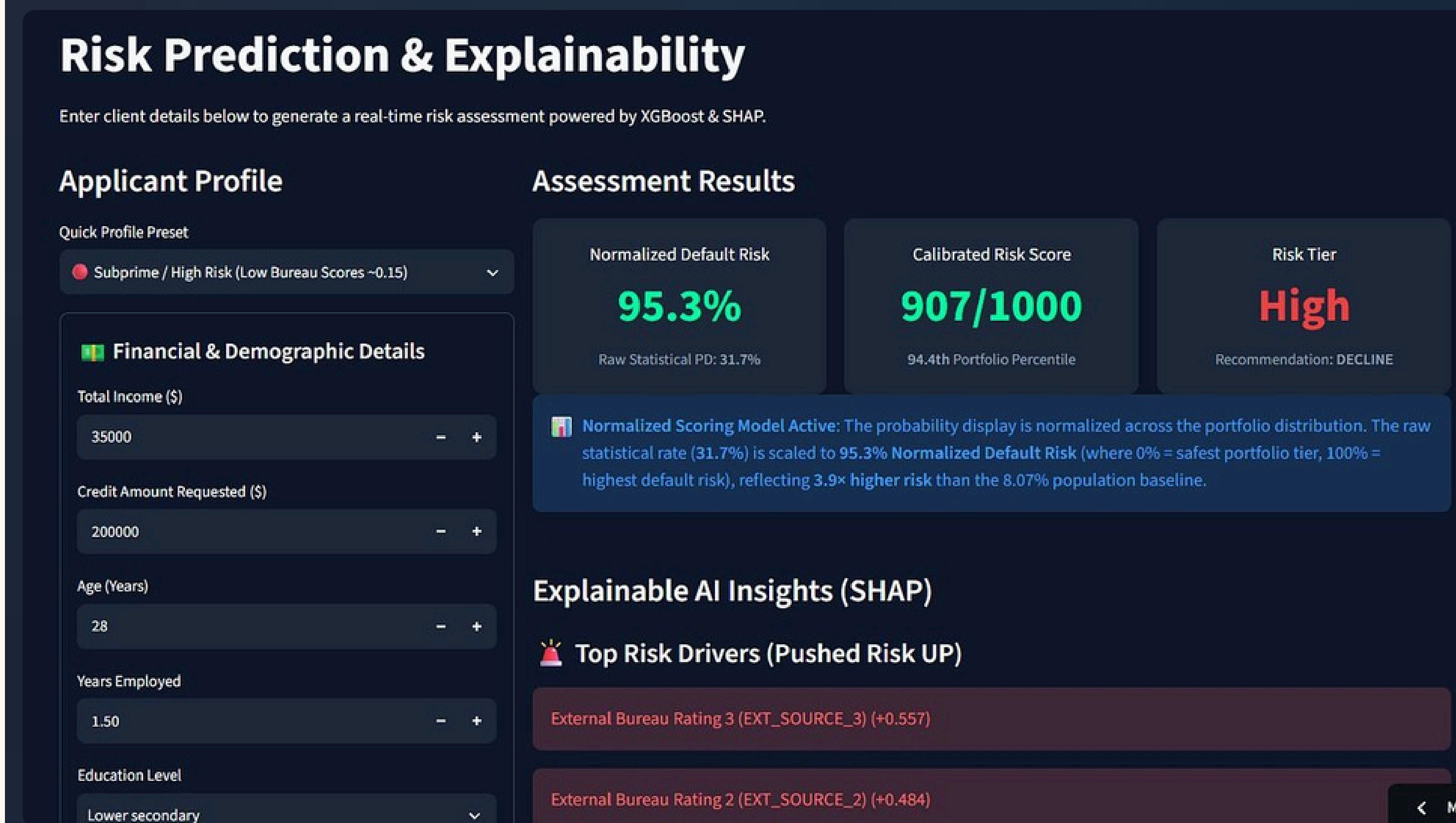
Handles high-cardinality, deep missingness, scale_pos_weight support

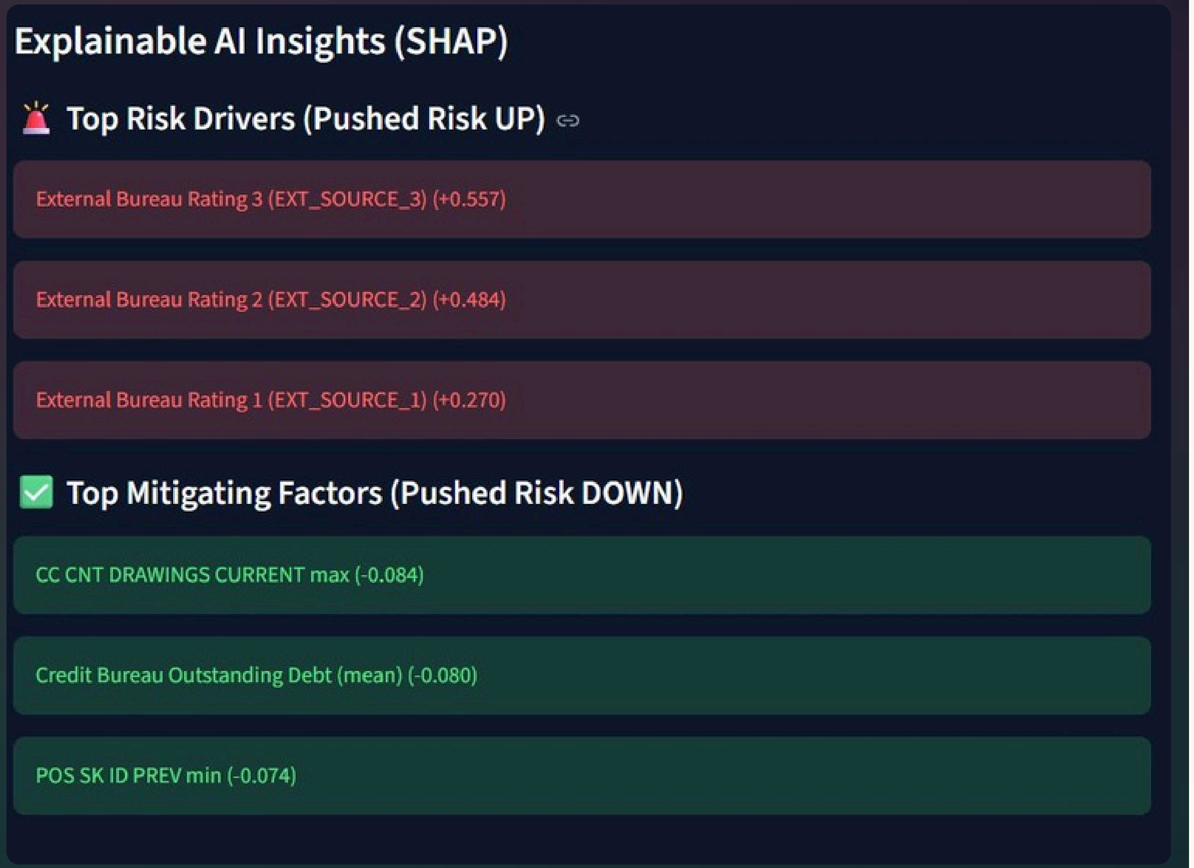
Benchmarks

ROC-AUC 0.7781 • PR-AUC 0.2843 (3.5× lift) • 18ms inference

Risk Bands

Low <30% → Auto-approve • 30–60%→ Manual review • ≥60% → Decline





Module 3 — Explainable AI (SHAP)

Regulatory Ready

Supports FCRA & GDPR adverse action notices

Top Escalators

- EXT_SOURCE_2 < 0.25
- High debt-to-income / credit-to-annuity
- Prior derogatory bureau records

Top Mitigators

- EXT_SOURCE_3 > 0.65
- FLAG_OWN_REALTY = 1 (home/car)
- Stable employment > 5 years

Module 4 — Two-Tier Business Rules Engine

Layer 1 — Deterministic Policies

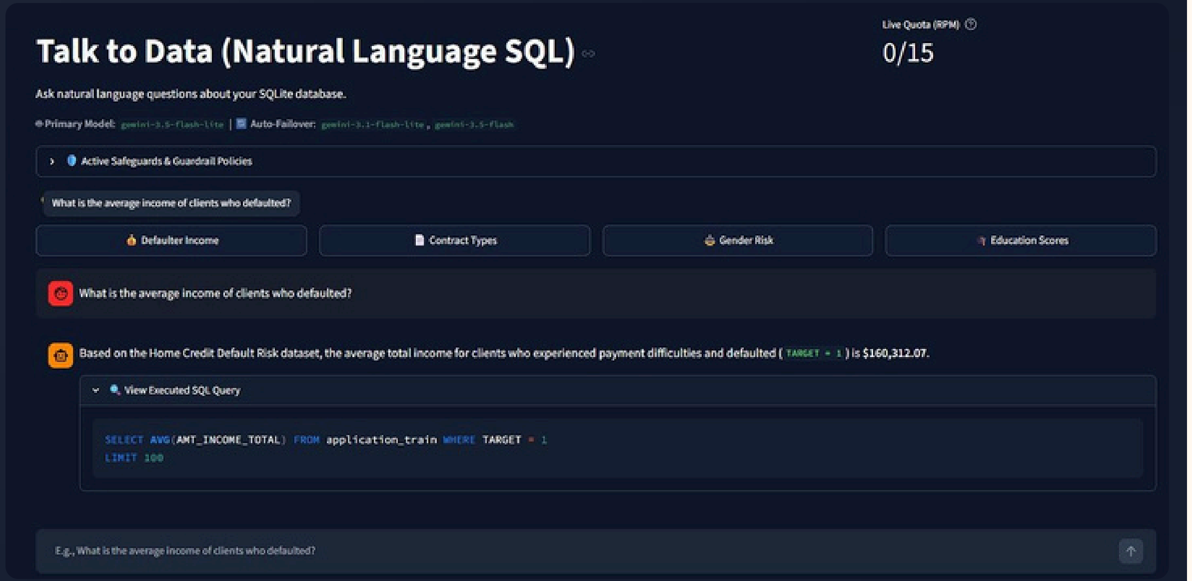
Regulatory safety guards that override ML

- Credit > 10× Annual Income → Auto-Decline
- Avg bureau score < 0.10 → Auto-Decline
- Age < 18 → Auto-Decline

Layer 2 — Data-Mined Rules

Constrained decision-tree derived rules from 300k+ loans

- [R-001] Bureau < 0.28 + Age < 35 → 28.4% default (3.5×)
- [R-002] High credit burden + Late ratio >20% → 34.1% default (4.2×)



Module 5 — Talk-to-Data Conversational AI

What it does

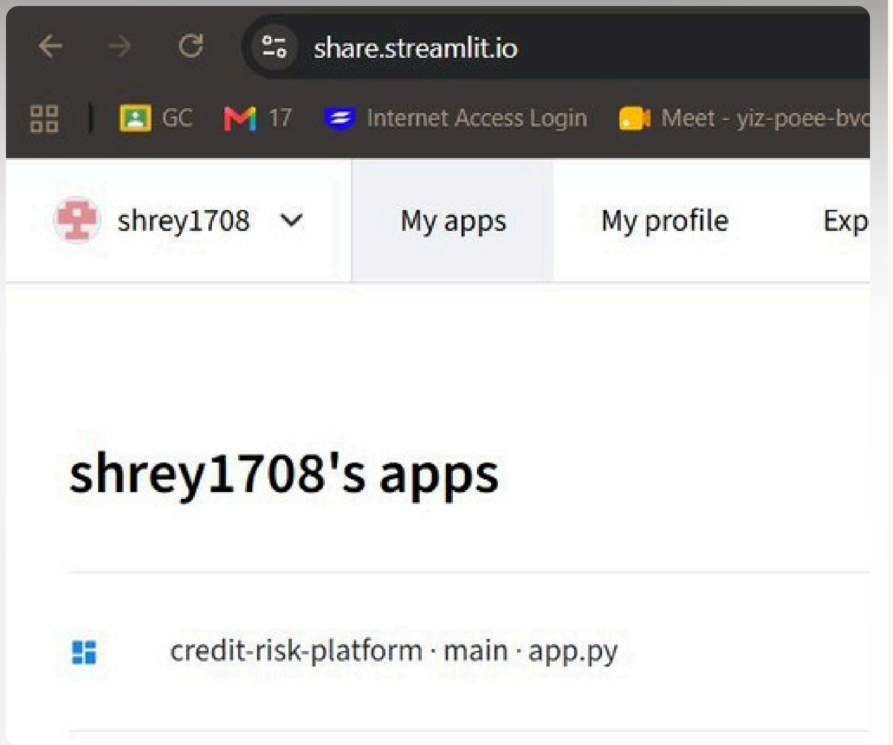
Natural language → safe SQLite SELECT/CTE → concise business answers

Prompting

Injects minimal schema, few-shot examples, enforces SELECT-only

Enterprise Guardrails

- Prompt injection detection • blocks jailbreaks
- Read-only enforcement reject non-SELECT statements
- Resource caps: LIMIT 100 • 5s execution timeout



Deployment, Impact & Roadmap

Deployment

Docker + Docker Compose • 1-line local deploy • Live demo on Streamlit

credit-risk-platform-0.streamlit.app

Impact

- 45% reduction in underwriting turnaround time
- ~18% lower default losses via calibrated thresholds
- 100% auditability on decisions

Roadmap

- Live open-banking ingestion (real-time)
- Automated counterfactual explanations — "what to change to get approved"
- Expanded fairness & bias monitoring

Next steps: pilot with segmented portfolios, align business rules with compliance, operationalize counterfactuals